

Vedant Saraswat

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EDUCATION

- University of California Irvine** - Master of Science in Computer Science Sept 2023 - Apr 2025
- Fully funded master's with **Graduate Teaching/Research Assistant** scholarships, focusing on Big Data and AI Applications.
 - **Courses** - Advanced ML, Reinforcement Learning, Efficient ML, Information Retrieval, Computational Geometry, Networks
- Indian Institute of Technology Kharagpur** - Bachelor of Technology (Honors) July 2017 - Apr 2021
- Minor Curriculum in **CS** - Programming & Data Structures, Algorithms, Social Computing, Image Processing, Probability, ML
 - Department Rank 4 and awarded **RK Bramha Cup** for the Best final year (Technology) and recognition letter from Director

EXPERIENCE

- Member of Technical Staff** - (Platform Engineering Team) - [Articul8 AI](#) July 2025 - Present
- Designed a distributed multi-agent orchestration platform (**AgentMesh**) using Rust and Python for domain-specific retrieval and reasoning, enabling agent-to-agent search, parallel execution, and dynamic model routing with built-in failure handling.
 - Architected a **scalable knowledge and ingestion pipeline** using Ray, ArangoDB, Milvus, Redis to support document/image/table extraction, semantic search, logical linking, and knowledge graph construction.
 - Worked on platform infrastructure and observability using Terraform, Kubernetes Helm charts, Karpenter for autoscaling, and monitoring stack (Prometheus, Grafana, Loki) across AWS/GCP, implementing GitOps workflows, CI/CD pipelines.
- Machine Learning Intern** - (Quality of Engineering Team) - [Viasat](#) June 2024 - Sep 2024
- Developed ML **forecasting models** on AWS to predict active network speed traffic patterns, optimizing bandwidth allocation.
 - Designed ETL pipelines with **SQL** in Snowflake for models ranging from regressors to **t-Transformers** getting RMSE under 6
- Data & Tech Fellow** - (Professor Tom Andriola) - [University of California Irvine](#) Sep 2023 - Mar 2024
- Developed a **Mixture of Experts LLM** chatbot from scratch to deliver student-oriented responses for diverse academic queries.
 - Built GPU-optimized **RAG** platform to automate the creation of sustainability dashboards based on AASHE data, with web integration via Streamlit to facilitate efficient querying and drive informed decision-making to enhance sustainability initiatives
- NLP Researcher** - (Dr Dan Roth) - [Cognitive Computation group](#) Aug 2023 - Sep 2024
- As part of Dr Dan Roth's Cognitive Computation group at the University of Pennsylvania, under the mentorship of Dr Vivek Gupta: Innovated Methods for **Tabular Natural Language Inference**.
- Software Development Engineer** - (Web & Games Team) - [Amazon MX Player](#) Jun 2021 - Aug 2023
- Orchestrated end-to-end creation of a **HLS Streaming Player** for various web OS with inbuilt DRM protection and **dynamic ad insertion**, utilizing VideoJs, Widevine, and serving seamless experiences for 175 million people.
 - Engineered full-stack user flows, covering **Authentication**, Subscriptions, **Push Notifications** and **Content Recommendation** from scratch utilizing TypeScript, React, Next.js, Spring, and an adaptive SageMaker-based recommender engine.
- Undergraduate Researcher** - (Professor Pawan Goyal - Thesis Project) - [CNERG Labs](#) Jun 2020 - Aug 2021
- Detected protein interaction in medical texts based on shortest **dependency knowledge graphs** and common sense encoders.
 - Researched suicidal personality prediction on extracted multimodal Facebook data using **T-LSTM** and **Deep BOSE** network
- Software Development Intern** - (Transportation Tech team) - [Amazon](#) Apr 2020 - Jun 2020
- Enhanced client onboarding in fleets by creating a **Spring** backend module to rectify errors, utilizing Postgres Database to manage **data disparities** between container and their Sable Database API calls on the frontend developed with JSP.
 - Formulated an algorithm to optimize **Transactions Per Second (TPS)** effectively reducing latency for the TCDA portal.
- Data Science Intern** - (IBUTUS team) - [Tech Mahindra](#) May 2019 - Jun 2019
- Optimized Model Performance on 30 million VoIP instances solving label imbalance and contributed to the sk-multilearn library.
 - Applied **cost-sensitive RL algorithms**, reducing SCUMBLE by 15% and enhancing macro-F1 score by 10.5% via ensemble.

PROJECTS

- EHR Agentic Chat** - An LLM agent augmented with tools & medical data, to solve complex tab. reasoning derived from EHR.
- Speculative Decoding** - Experimented optimization pipeline to accelerate language model inference, achieving 3x speedups.
- UCI Search** - Engineered IR Search Engine on CS Database with PageRank, n-gram indexing, RAG, and RLHF/PPO techniques.
- Smart Chat Assistant** - Designed an AB Inbev Business and Customer-centric proactive Chatbot with interactive AI webhooks

SKILLS

Languages/Scripts	C, C++, Python, Go, Java, Javascript, Typescript, Rust, Octave, Matlab, HTML, Bash, SQL
Libraries	OpenCv, Numpy, Pandas, React, Next.js, Ray, FastAPI, NLTK, Flask, Django, Pytorch, Spring, Spark
Infra/Databases	Kubernetes, Terraform, Docker, AWS, GCP, Azure, MongoDB, Arango, Milvus, Redis, Prometheus, Git